

Curriculum Vitae

Baixin Wan

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Education

Southeast University **B.E. in Computer Science and Technology** **Aug. 2023 – Expected Jun. 2027**

- **Average Score: 87.96**
- Outstanding Student Leader (2023–2024), Merit Student (2024–2025)
- National Encouragement Scholarship (2024–2025)
- Second Prize in Lanqiao Cup (Provincial Level, Group A), 2025

Publications & Manuscripts

Accepted B. Wan^{*,†}, F. Shan, and J. Huang, “From Docking Station to Docking Station: Completing Tasks in Minimum Time by Cooperative UAV Fleets,” *IEEE International Conference on Parallel and Distributed Systems*, 2025 .

In Preparation B. Wan^{*,†} et al., “Working Title: Heterogeneous Cooperative UAV Swarm Optimization with Minimum Time Objective,” intended for submission to *Journal of Systems Architecture (JSA)*, 2026 .

* Sole First Author † Sole Student Author

Research Experience

Core Member **Algorithm Design for Parallel Scheduling on Heterogeneous UAVs** **2025.06–Present**
Southeast University *advised by Prof. Feng Shan*

- Proposed a DP algorithm that computes the optimal solution for a multi-UAV cooperative scheduling problem. The method minimizes completion time under sequential execution and multi-UAV collaboration constraints, and further designed scalable heuristics for large-scale instances.
- Extended the scheduling framework to heterogeneous UAV fleets, developing corresponding formulations, heuristic algorithms. On large-scale instances, the proposed method achieved a 10% optimization compared to the baseline algorithm.
- In this project, I was primarily responsible for algorithm design, code implementation, and manuscript preparation.

Research Intern **Heterogeneous Acceleration for RAG with High-cost Update** **2025.12–Present**
Peking University *advised by Prof. Xiang Chen*

- Surveyed and reproduced recent Agent frameworks involving RAG operation such as ReMe and Dynamic Cheatsheet, and summarized heterogeneous execution inefficiencies in current agent scheduling, where disambiguation, graph reconstruction, and index updates experience severe resource contention with the retrieval and generation phases.
- Surveyed parallelization strategies in RAG pipelines such as HedraRAG, HippoRAG, G-Memory and Pancake, and identified a preliminary acceleration direction based on speculative execution and decoupling of independent components.
- Manuscript in preparation

Research Intern **Image Generation with LLaMA and DiT Models** **2025.11–2026.01**
Nanjing University *advised by Prof. Qi Fan*

- Reproduced representative image generation models on CIFAR-10, including a LLaMA-based autoregressive sequence model and a DiT-based diffusion model, to study both token-based and diffusion-based generation paradigms.
- Explored single-step generation through knowledge distillation by training student models to mimic the behavior of iterative generators and evaluating their generation quality.
- Conducted preliminary experiments on speculative generation by training lightweight draft models.

Open-source & Course Projects

- DS Expr** Data structure experiments involving external sorting and cache read/write operations. (*Rust + Python*)
- Rust Unsafe** Educational experiments on Rust unsafe programming and FFI (Foreign Function Interface), implementing reciprocal calls between Rust-based key-value storage and C. (*Rust*)
- Rust Interp** Rust implementation of the “Let’s Build A Simple Interpreter” tutorial from Ruslan’s blog. (*Rust*)
- VCampus** Served as the team leader to design and implement a multi-module virtual campus management system. (*Java + Maven + MySQL*)

Skills

- Academic** Proficient in $\text{\LaTeX}$ and Markdown; familiar with Google Scholar, dblp, IEEE Xplore; skilled in Git, Vim, Docker, and design software (Photoshop, Illustrator).
- Frontend** Proficient in HTML/TS/CSS; familiar with Vue.js.
- Backend** Proficient in Rust and C++; familiar with Rust unsafe development and FFI; familiar with Python.